UPPSALA UNIVERSITET

Department of Information Technology

FRONT SHEET FOR EXAMS	DATE: 2024-06-07
Course name 1RT485 - Introduction to Computer Control Systems	
Your exam code | | | | | | – | | | | | – | | |	
Semester when you were first registered to the course:	Programme:
Time for submitting the exam:	Table number:

INSTRUCTIONS

Please check that you have the correct exam!

This sheet should always be submitted, even if you haven't solved any of the exam problems.

Each solution should be written on a separate paper.

Write your exam code on each paper.

Please use only *one side* of the papers and do not use a pencil with red colour.

Sort the solutions in question order, with question 1 first, before you submit them.

No.	Solved problems (mark with X)	Points earned	Comments from the teacher
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

∑
☐ Exam with bonus points: Grade is not shown.[1]
5 ≥ ☐ 4 ≥ ☐ 3 ≥ ☐
Exam grade:

1 The final result (points including bonus points and grade) can be found in Ladok after certification.

Introduction to Computer Control Systems, 5 credits, 1RT485

Date: 2024-06-07

Teacher on duty: Fabio Bonassi

Number of problems: 5

Allowed aid: A non-programmable calculator and mathematical handbooks

Preliminary grades:	grade 3	15 points
	grade 4	21 points
	grade 5	26 points

Some general instructions and information:

- Your solutions should be given in English.
- Write only on one side of the paper.
- Write your exam code and page number on all pages.
- Do not use a red pen.
- Use separate sheets of paper for the different problems (i.e. the numbered problems, 1–5).

With the exception of Problem 1, ***all your answers must be clearly motivated!*** *A correct but unmotivated answer will score reduced points!*

Best of luck!

Formula sheet

Laplace transform table

Table 1: Basic Laplace transforms

$f(t)$	$F(s)$	$f(t)$	$F(s)$
unit impulse — imp(t)	1	$e^{-at}\sin(bt)$	$\frac{b}{(s+a)^2+b^2}$
unit step — step(t)	$\frac{1}{s}$	$e^{-at}\cos(bt)$	$\frac{s+a}{(s+a)^2+b^2}$
unit ramp — ramp(t)	$\frac{1}{s^2}$	$\sinh(bt)$	$\frac{b}{s^2-b^2}$
t	$\frac{1}{s^2}$	$\cosh(bt)$	$\frac{s}{s^2-b^2}$
t^n	$\frac{n!}{s^{n+1}}$	$\frac{1}{2b}t\sin(bt)$	$\frac{s}{(s^2+b^2)^2}$
e^{-at}	$\frac{1}{s+a}$	$t\cos(bt)$	$\frac{s^2-b^2}{(s^2+b^2)^2}$
$\frac{1}{a}(1-e^{-at})$	$\frac{1}{s(s+a)}$	$\frac{\cos(bt)-\cos(at)}{a^2-b^2}; (a^2 \neq b^2)$	$\frac{s}{(s^2+a^2)(s^2+b^2)}$
$\frac{1}{(n-1)!}t^{n-1}e^{-at}; (n=1,2,3...)$	$\frac{1}{(s+a)^n}$	$\frac{\sin(at)+at\cos(at)}{2a}$	$\frac{s^2}{(s^2+a^2)^2}$
$\sin(bt)$	$\frac{b}{s^2+b^2}$		
$\cos(bt)$	$\frac{s}{s^2+b^2}$		

Table 2: Properties of Laplace Transforms

$\mathcal{L}[af(t)] = aF(s)$	$\mathcal{L}[tf(t)] = -\frac{dF(s)}{ds}$
$\mathcal{L}[f_1(t)+f_2(t)] = F_1(s)+F_2(s)$	$\mathcal{L}\left[t^2 f(t)\right] = \frac{d^2}{ds^2}F(s)$
$\mathcal{L}\left[\frac{d}{dt}f(t)\right] = sF(s)-f(0)$	$\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n}{ds^n}F(s), \quad n \in \mathbb{N}$
$\mathcal{L}\left[\frac{d^2}{dt^2}f(t)\right] = s^2F(s)-sf(0)-f'(0)$	$\mathcal{L}\left[f\left(\frac{t}{a}\right)\right] = aF(as)$
$\mathcal{L}\left[\int f(t)\,dt\right] = \frac{F(s)}{s}+\frac{1}{s}\left[\int f(t)\,dt\right]_{t=0}$	$\mathcal{L}\left[\int_0^t f_1(t-\tau)f(t)d\tau\right] = F_1(s)F_2(s)$
$\mathcal{L}[f(t-a)] = e^{-as}F(s)$	$\mathcal{L}[e^{-at}f(t)] = F(s+a)$

Open-loop and sensitivity functions

$$G_o(s) = G(s)F_y(s), \qquad S(s) = \frac{1}{1+G_o(s)}, \qquad T(s) = 1-S(s)$$

State-space forms and transfer function relations

- State-space form and transfer function

$$\begin{aligned} \dot{x} &= Ax+Bu \\ y &= Cx+Du \end{aligned} \quad \Rightarrow \quad \boxed{G(s) = C(sI-A)^{-1}B+D}$$

- Associated matrices

$$\mathcal{S} = \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix} \qquad \mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

- LTI system with transfer function

$$\boxed{G(s) = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_n}{s^n + a_1 s^{n-1} + \cdots + a_n}}$$

  i) Observable canonical form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -a_1 & 1 & 0 & \cdots & 0 \\ -a_2 & 0 & 1 & \cdots & 0 \\ -a_3 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ -a_n & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} b_1 - a_1 b_0 \\ b_2 - a_2 b_0 \\ b_3 - a_3 b_0 \\ \vdots \\ b_n - a_n b_0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b_0 u$$

  ii) Controllable canonical form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -a_1 & -a_2 & -a_3 & \cdots & -a_n \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & 0 & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} b_1 - a_1 b_0 & b_2 - a_2 b_0 & \cdots & b_n - a_n b_0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b_0 u$$

- Solution to state-space equation

$$\dot{x}(t) = Ax(t) + Bu(t), \quad x(0) = x_0$$

  can be written as

$$\boxed{x(t) = e^{At} x_0 + \int_0^t e^{A\tau} B u(t-\tau) d\tau}$$

- Observer system

$$\boxed{\dot{\hat{x}} = A\hat{x} + Bu + K(y - C\hat{x})}$$

Feedback control structures

General linear feedback in Laplace form:

$$\boxed{U(s) = F_r(s)R(s) - F_y(s)Y(s)}$$

Common control structures in this form.

- PID controller:

$$F_y(s) = F_r(s) = F(s) = K_p + \frac{K_i}{s} + K_d s,$$

  where $K_p, K_i, K_d \geq 0$

- State-feedback controller with observer:

$$\begin{aligned} F_r(s) &= \left(1 - L(sI - A + KC + BL)^{-1}B\right)\ell_0 \\ F_y(s) &= L(sI - A + KC + BL)^{-1}K \end{aligned}$$

Bode's phase diagram

Consider the loop gain $G_o(s) = \mu \frac{\prod_{k=1}^{n_z}(1-\tau_k s)}{s^g \prod_{k=1}^{n_p}(1-T_k s)}$, where for simplicity $\mu > 0$. The phase of $G(i\omega)$ at a given frequency is

$$\arg\left(G_o(i\omega)\right) = -g\,90^\circ + \left[\sum_{k=1}^{n_z} \arctan\left(\tau_k \omega\right) - \sum_{k=1}^{n_p} \arctan\left(T_k \omega\right)\right] \frac{180^\circ}{\pi}$$

Routh–Hurwitz stability criterion

Let $\Phi(s) = a_0 s^n + b_0 s^{n-1} + a_1 s^{n-2} + ...$ be the characteristic polynomial.

$$\begin{array}{ccc} a_0 & a_1 & ... \\ b_0 & b_1 & ... \\ \hline c_0 & c_1 & ... \\ \vdots & \ddots & \end{array}$$

A generic element of the third row, c_k, can be computed as

$$c_k = \frac{b_0 a_{k+1} - a_0 b_{k+1}}{b_0}$$

Discrete-time state-space forms

A continuous time system with zero-order-hold input signal and sample period T can be written in discrete-time as:

$$\boxed{\begin{aligned} x_k &= Fx_k + Gu_k \\ y_k &= Hx_k \end{aligned}}$$

where

$$
\begin{aligned}
F &= e^{AT} \\
G &= \int_{\tau=0}^{T} e^{A\tau} d\tau B = \lceil \text{if } A^{-1} \text{ exists} \rceil = A^{-1}(e^{AT} - I)B \\
H &= C
\end{aligned}
$$

Discrete-time state-space forms

A continuous time system with zero-order-hold input signal and sample period T can be written in discrete-time as:

$$
\boxed{\begin{aligned} x(k+1) &= Fx(k) + Gu(k) \\ y(k) &= Hx(k) \end{aligned}}
$$

where

$$
\begin{aligned}
F &= e^{AT} \\
G &= \int_{\tau=0}^{T} e^{A\tau} d\tau B = \lceil \text{if } A^{-1} \text{ exists} \rceil = A^{-1}(e^{AT} - I)B \\
H &= C
\end{aligned}
$$

Bilinear transform formula ($\alpha \in \{0, \frac{1}{2}, 1\}$)

$$
s = \frac{1}{T} \frac{1 - q^{-1}}{\alpha + (1 - \alpha)q^{-1}}
$$

Problem 1: basic questions (6/30)

Answer only *'true'* or *'false'*. Each correct answer gives 1 point, each wrong answer gives -1 point. Minimum total points for Part A and B is 0 , respectively.

Part A

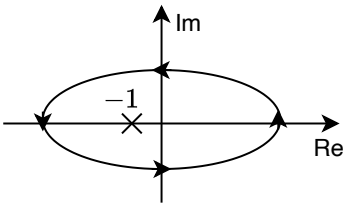
Note: Write 'skip' if your total home assignment score ≥ 8

i) The system

$$G(s) = \frac{s+1}{(s^2 - s + 5)(s+5)}$$

is input-output stable.

ii) The system $G(s)$ has exactly 1 pole with positive real part. You wish to control this system with the proportional controller $F(s) = 10$. Knowing the Nyquist diagram of the loop gain $G_o(s)$, reported below, you can conclude that the closed-loop is stable.



iii) In PID controllers, the derivative part allows to reduce oscillations and to remove the steady-state error.

(3 p)

Part B

Note: Write 'skip' if your total home assignment score ≥ 13

i) To reduce the closed-loop resonance peak one needs to increase the phase margin.

ii) A transfer function $G(s)$ admits only one minimal state-space realization.

iii) Some controller $F(s)$, which ensures a closed-loop bandwidth $\omega_B = 10$ rad/s, needs to be discretized so that it can be implemented on a computer. Discretizing $F(s)$ with a sampling frequency $\omega_s = 2 \cdot \omega_B = 20$ rad/s is enough to achieve similar closed-loop performance.

(3 p)

Proposed solution to problem 1

Part A

i) False

ii) True

iii) False

Part B

i) True

ii) False

iii) False

Problem 2 (6/30)

A mechanical system is described by the transfer function

$$G(s) = \frac{s-1}{s^2+s+5}$$

The task is to control it with a proportional controller $F(s) = K$.

a) For which values of K is the closed-loop stable?

(2 p)

b) Compute the value of K such that the closed-loop poles are real and coinciding.

(2 p)

c) Given the proportional controller designed at point **b)**, what is the steady-state error when the reference is $r(t) = 1 \cdot \text{step}(t)$?
Note: use $K = 2$ if and only if you didn't solve point **b**)

(2 p)

Proposed solution to problem 2

a) The open-loop transfer function is $G_o(s) = K\frac{s-1}{s^2+s+5}$, while the closed-loop is $G_c(s) = \frac{G_o(s)}{1+G_o(s)} = \frac{K(s-1)}{s^2+(K+1)s+5-K}$.
Applying, for example, the Routh criterion, we get

$$\begin{cases} K+1>0 \\ 5-K>0 \end{cases}$$

That is $-1 < K < 5$.

b) The closed-loop poles are the roots of the denominator of $G_c(s)$.

$$s^2 + (K+1) + 5 - K = 0 \quad \Longrightarrow \; s = -\frac{K+1}{2} + \frac{1}{2}\sqrt{(K+1)^2 - 4(5-K)}$$

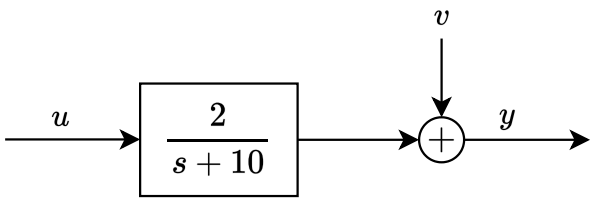
The poles are real and coinciding if and only if the term under the square root is exactly equal to zero.

$$(K+1)^2 - 4(5-K) = 0 \implies K^2 + 6K - 19 = 0 \implies K = \frac{-6 \pm \sqrt{112}}{2}.$$

The only acceptable solution is $K = \frac{-6+\sqrt{112}}{2} \approx 2.29$. The other solution is indeed < -1.

c) When the reference is a unity step ($R(s) = \frac{1}{s}$), the closed-loop output is $Y(s) = G_c(s)R(s)$. Then, according to the Final Value theorem,

$$\begin{aligned} e(\infty) &= \lim_{s\to 0} s\left(Y(s) - R(s)\right) = \lim_{s\to 0} s(G_c(s)-1)\frac{1}{s} \\ &= \lim_{s\to 0}\left(\frac{2.29(s-1)}{s^2+3.29s+2.71} - 1\right)\frac{s}{s} = -\frac{2.29}{2.71} - 1 \approx -1.84. \end{aligned}$$



Problem 3 (7/30)

When you join ACME Inc., you are tasked with the problem of designing a control system for a device described by the following transfer function:

$$G(s) = \frac{2}{s+10}$$

The system is also affected by a sinusoidal disturbance $v(t) = \sin(\omega t)$ (see the figure) with frequency $\omega = 1$ rad/s. You want to kickstart the project by designing a proportional controller $F(s) = K$, where $K > 0$.

a) Assume that the disturbance $v(t)$ is not measurable. For what values of K is the closed-loop system able to attenuate the disturbance by a factor of (at least) 10?

(3.5 p)

b) Assume now that the disturbance $v(t)$ is instead measured. Design a feed-forward compensator that compensates for this disturbance (draw the diagram and compute the compensator explicitly, making sure it can be implemented).

(3.5 p)

Proposed solution to problem 3

a) The Laplace transform of the output is

$$Y(s) = G_c(s)R(s) + S(s)D(s).$$

Therefore, in order to have an attenuation factor of 10, $|S(i\omega)| \leq \frac{1}{10}$.

$$|S(i\omega)| = \left|\frac{1}{1+G_o(i\omega)}\right| = \left|\frac{s+10}{2K+s+10}\right|_{s=i\omega} = \frac{\sqrt{1^2+10^2}}{\sqrt{1^2+(2K+10)^2}}$$

$|S(i\omega)| \leq 0.1$ if

$$0.1\sqrt{1^2+(2K+10)^2} \geq \sqrt{1^2+10^2}.$$

Squaring both sides

$$0.01 + 0.01(2K+10)^2 > 101$$

Since $K > 0$, the solution is

$$2K + 10 > \frac{\sqrt{100.99}}{0.1} \implies \boxed{\text{K} > 45}$$

b) Let's denote by $F_f(s)$ the disturbance compensator. The resulting control law is

$$U(s) = F(s)E(s) + F_f(s)D(s).$$

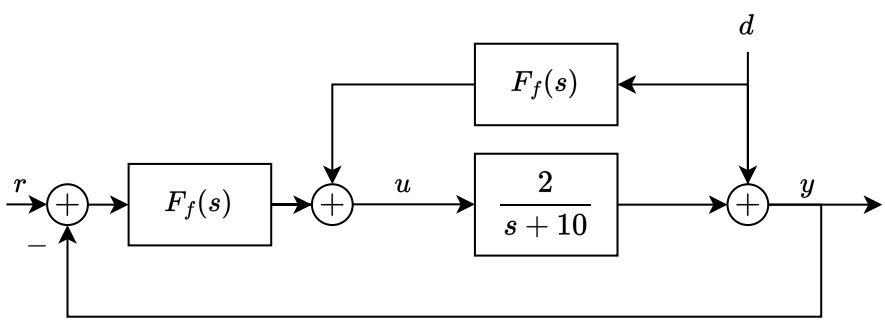
The ideal compensator is

$$F_f^\star(s) = -\frac{H(s)}{G(s)} = -\frac{1}{\frac{2}{s+10}} = -\frac{s+10}{2}.$$

This compensator cannot be implemented because it has more zeros than poles. Therefore, we just take the static disturbance compensator:

$$F_f(s) = F_f^\star(0) = -\frac{10}{2} = -5.$$

The resulting scheme is reported in the figure below.



Problem 4 (6/30)

You are given the state-space system

$$\dot{x} = \begin{bmatrix} -1 & 1 \\ -1 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$
$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} x$$

a) Design a state-feedback controller $u = -Lx + \ell_0 r$ such that the closed-loop eigenvalues are in -5 and -9. Find ℓ_0 so that the steady-state tracking error is zero for $r(t) = \text{step}(t)$.

(3 p)

b) Design a state observer so that the state estimation error has both the eigenvalues in -25.

(3 p)

Proposed solution to problem 4

a) Firstly, we need to verify that the system is controllable.

$$\mathcal{S} = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & -3 \end{bmatrix} \to \det(\mathcal{S}) \neq 0 \to \text{ok!}$$

Then, we compute the characteristic polynomial

$$\varphi(\lambda) = \det(\lambda I - (A + BL)) = \det \begin{bmatrix} \lambda + \ell_1 + 1 & -1 + \ell_2 \\ 1 + \ell_2 & \lambda + 2 + \ell_2 \end{bmatrix} = \lambda^2 + (3 + \ell_1 + \ell_2)\lambda + 3 + 3\ell_1$$

Comparing it to the desired characteristic polynomial $(\lambda + 5)(\lambda + 9) = \lambda^2 + 14\lambda + 45$ we get

$$\begin{cases} 3 + \ell_1 + \ell_2 = 14 \\ 3 + 3\ell_1 = 45 \end{cases} \implies L = \begin{bmatrix} 14 & -3 \end{bmatrix}$$

Finally, we compute ℓ_0 so that the static gain is 1. From the lecture slides,

$$\ell_0 = \frac{1}{C(-A + BL)^{-1}B} = \frac{1}{\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 15 & -4 \\ 15 & -1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 1 \end{bmatrix}} = \frac{1}{\frac{1}{45} \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 4 \\ -15 & 15 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}} = 15$$

The state-feedback controller will therefore be $u = -\begin{bmatrix} 14 & -3 \end{bmatrix} x + 15r$.

b) We first check the observability of the system.

$$\mathcal{O} = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \to \det(\mathcal{O}) \neq 0 \to \text{ok!}$$

The characteristic polynomial of the state estimation error is

$$\varphi(\lambda) = \det(\lambda I - (A - KC)) = \det \begin{bmatrix} \lambda + 1 + \kappa_1 & -1 \\ 1 + \kappa_2 & \lambda + 2 \end{bmatrix} = \lambda^2 + (\kappa_1 + 3)\lambda + 2\kappa_1 + 3 + \kappa_2$$

Comparing it to the desired characteristic polynomial $(\lambda + 25)^2 = \lambda^2 + 50\lambda + 625$ we get

$$\begin{cases} \kappa_1 + 3 = 50 \\ 2\kappa_1 + 3 + \kappa_2 = 625 \end{cases} \implies K = \begin{bmatrix} 47 \\ 528 \end{bmatrix}$$

The state observer is

$$\dot{\hat{x}} = A\hat{x} + Bu + K(y - C\hat{x}) = \begin{bmatrix} -48 & 1 \\ -529 & -2 \end{bmatrix} \hat{x} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u + \begin{bmatrix} 47 \\ 528 \end{bmatrix} y$$

Problem 5 (5/30)

A colleague of yours has to design a control system for a mechanical system. The goal is to have static accuracy (steady-state output error as small as possible) and good damping (no overshoots in the closed-loop response).

a) He is torn between two different feedback controllers:

$$\text{Option A:} \quad F(s) = 4\,\frac{s+0.4}{s}$$
$$\text{Option B:} \quad F(s) = 2\,\frac{s+4}{s+0.05}$$

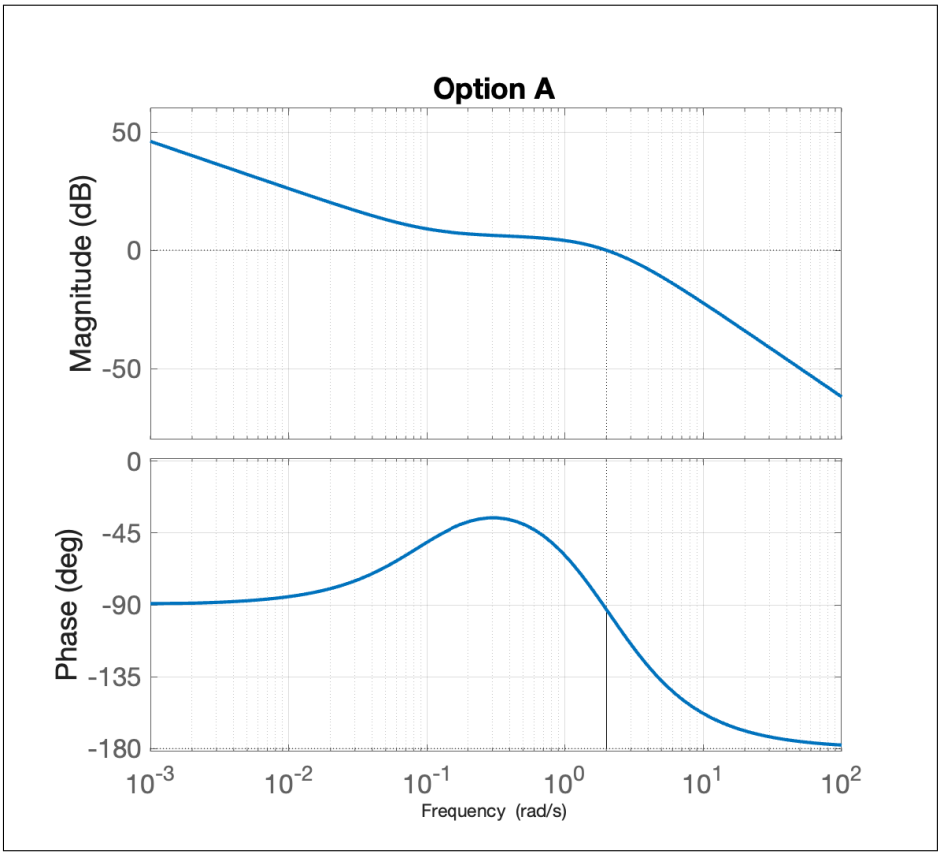
To decide, he shows you the two bode diagrams of the open-loop transfer function $G_o(s)$ in the two cases (see the figure in the next page). Which of the two controllers would you recommend? Justify your answer.

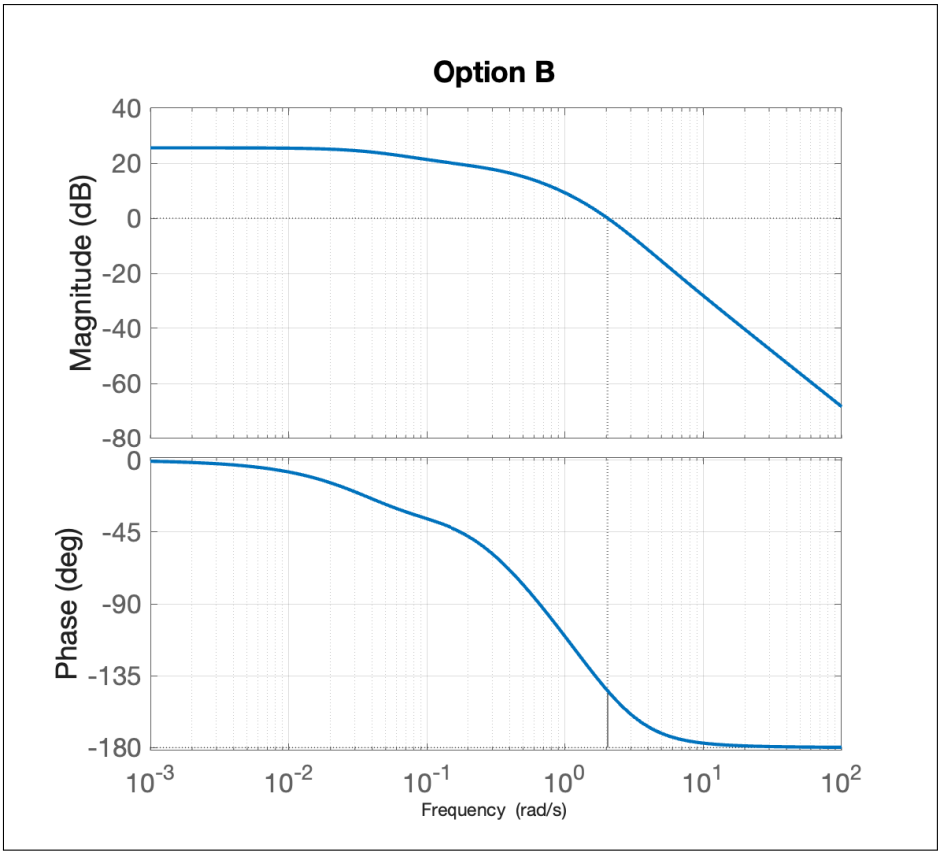
(2.5 p)

b) Discretize the control law $U(s) = F(s)E(s)$, where $F(s)$ is the controller that you selected at point **a)**. Use either the trapezoidal method (Tustin) or with Backward Euler, with sampling time $T = 0.1$ seconds. The resulting discrete-time control law should be in the form

$$u_k = a_1 u_{k-1} + b_0 e_k + b_1 e_{k-1}$$

(2.5 p)


Option A
Magnitude (dB)
50
0
-50
Phase (deg)
0
-45
-90
-135
-180
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Frequency (rad/s)


Option B
Magnitude (dB)
40
20
0
-20
-40
-60
-80
Phase (deg)
0
-45
-90
-135
-180
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Frequency (rad/s)

Proposed solution to problem 5

a) Option A is the one that achieves the desidered goals, so it is the best option.

- $\lim_{s\to 0^+} G_o(s) = +\infty$, meaning that the steady-state output error is zero.
- The phase margin, which is related to the damping of the closed-loop response, is large ($\varphi_m = +90°$). The closed-loop response is well damped.

On the other hand, Option B has a quite-small steady-state gain ($G_o(0) <$ 20dB), which implies an static error of $\frac{0.1}{1+0.1} \approx 9\%$, and a small phase margin ($\varphi_m < 40°$), which means that the closed-loop response would be oscillatory.

b) Given the control law $U(s) = 4\,\frac{s+0.4}{s}$, we can apply the tustin formula $s \leftarrow \frac{2}{T}\frac{1-q^{-1}}{1+q^{-1}} = 20\frac{1-q^{-1}}{1+q^{-1}}$. Then,

$$U(q) = 4\frac{20\frac{1-q^{-1}}{1+q^{-1}} + 0.4}{20\frac{1-q^{-1}}{1+q^{-1}}}E(q) = \frac{80 - 80q^{-1} + 1.6 + 1.6q^{-1}}{20 - 20q^{-1}}E(q).$$

Hence, the discrete-time control law is

$$20(1-q^{-1})u_k = (81.6 - 78.4q^{-1})e_k \quad \rightarrow \quad \boxed{u_k = 1u_{k-1} + \frac{81.6}{20}e_k - \frac{78.4}{20}e_{k-1}}$$